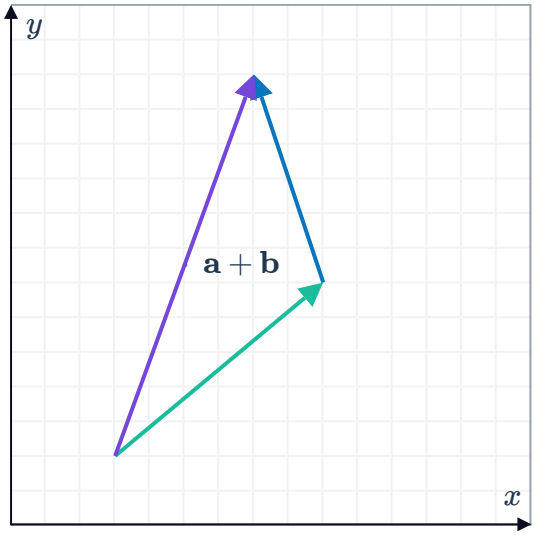
Geometric addition and subtraction of vectors (Extended)

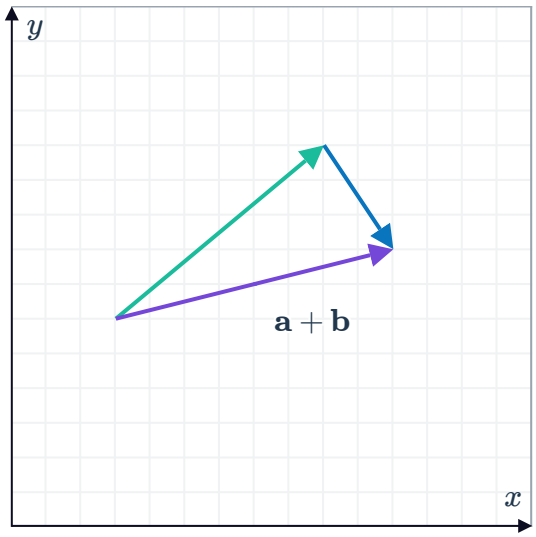
1 **c**

2

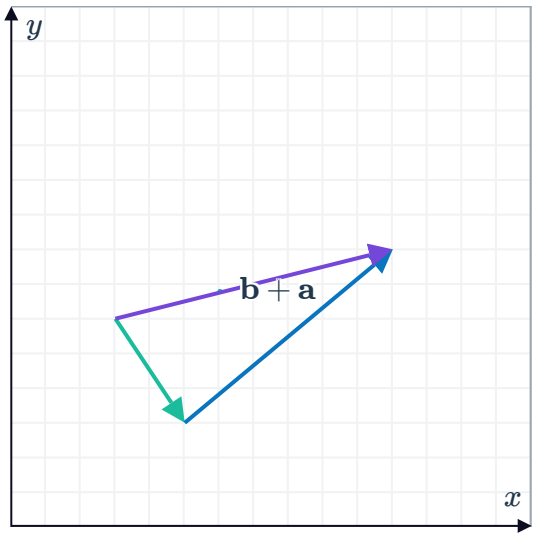


3

a



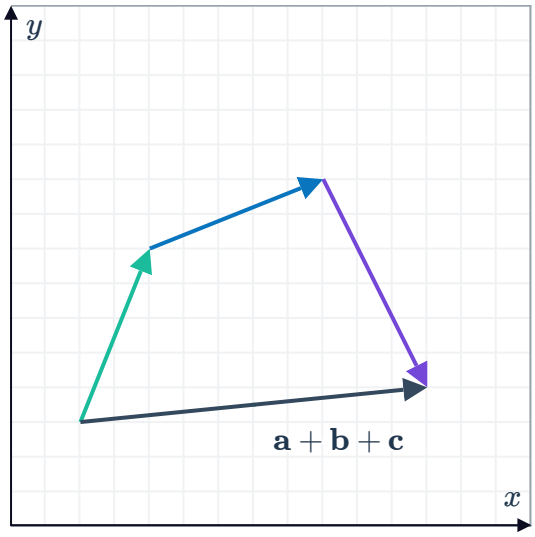
b



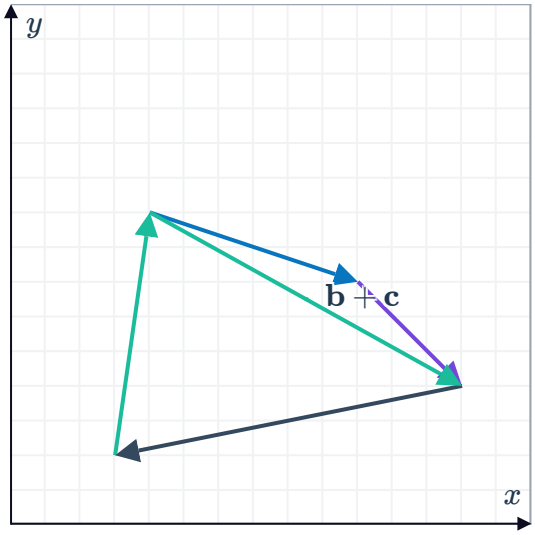
c Yes

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4



5



Algebraic addition and subtraction of vectors

6 $(13, 1)$

7

a

i $\begin{pmatrix} 12 \\ 11 \end{pmatrix}$ ii $\begin{pmatrix} 15 \\ 7 \end{pmatrix}$ iii $\begin{pmatrix} 19 \\ 16 \end{pmatrix}$

b

i $\begin{pmatrix} -10 \\ -1 \end{pmatrix}$ ii $\begin{pmatrix} -2 \\ -7 \end{pmatrix}$ iii $\begin{pmatrix} -8 \\ 0 \end{pmatrix}$

c

i $\begin{pmatrix} 12.1 \\ -3.9 \end{pmatrix}$ ii $\begin{pmatrix} 2.3 \\ -1.2 \end{pmatrix}$ iii $\begin{pmatrix} 7 \\ -7.4 \end{pmatrix}$

d

i $\begin{pmatrix} \frac{13}{30} \\ -\frac{1}{5} \end{pmatrix}$ ii $\begin{pmatrix} \frac{11}{10} \\ -\frac{11}{5} \end{pmatrix}$ iii $\begin{pmatrix} \frac{29}{15} \\ -1 \end{pmatrix}$

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Scalar multiplication of vectors

8 Let $\mathbf{a} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$, find:

a $2\mathbf{a}$ **b** $5\mathbf{a}$ **c** $-1\mathbf{a}$ **d** $-6\mathbf{a}$

9 Let $\mathbf{a} = \begin{pmatrix} 10 \\ 16 \end{pmatrix}$, find:

a $3\mathbf{a}$ **b** $-10\mathbf{a}$ **c** $\frac{1}{2}\mathbf{a}$ **d** $-\frac{1}{4}\mathbf{a}$

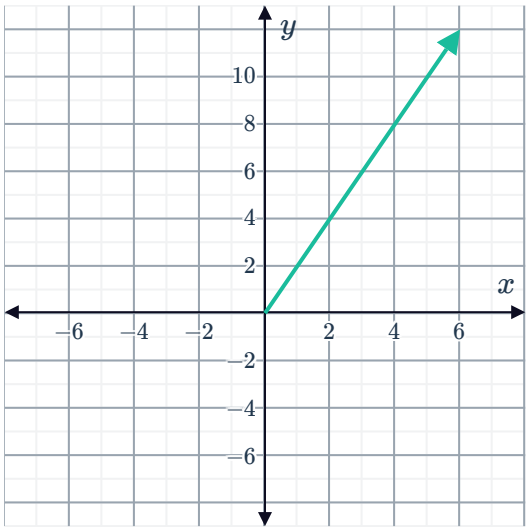
10 Let $\mathbf{a} = \begin{pmatrix} -4 \\ 9 \end{pmatrix}$, find:

a $3\mathbf{a}$ **b** $-2\mathbf{a}$ **c** $\frac{1}{2}\mathbf{a}$ **d** $-100\mathbf{a}$

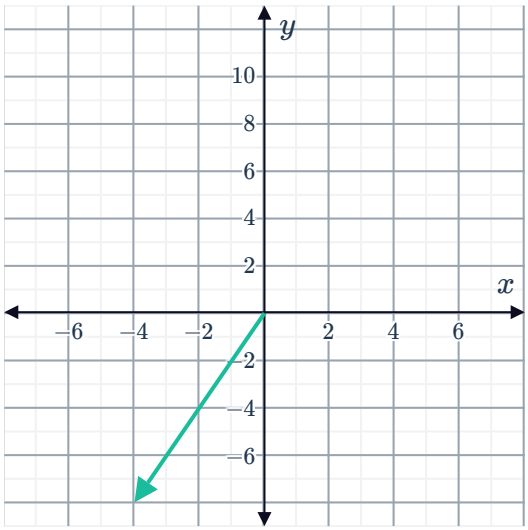
Scalar multiplication of vectors (Extended)

11

a

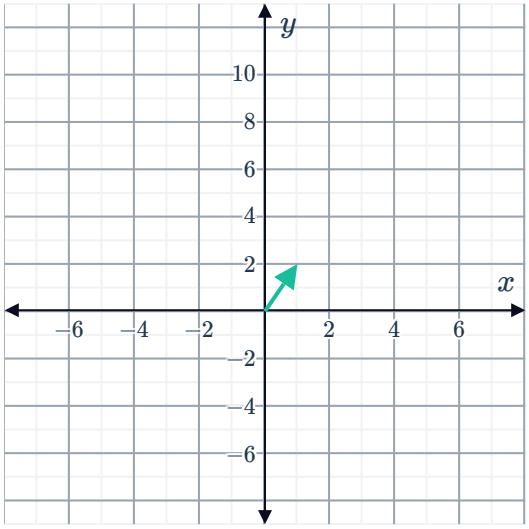


b



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c



12

a $\begin{pmatrix}-1\\11\end{pmatrix}$ **b** $\begin{pmatrix}9\\15\end{pmatrix}$ **c** $\begin{pmatrix}-16\\24\end{pmatrix}$ **d** $\begin{pmatrix}19\\19\end{pmatrix}$

e $\begin{pmatrix}-24\\-2\end{pmatrix}$

13

a **b** **b** **c** **c** **d** **d** **c**

14

a $\begin{pmatrix}6\\6\end{pmatrix}$ **b** $\begin{pmatrix}-12\\16\end{pmatrix}$ **c** $\begin{pmatrix}-6\\-6\end{pmatrix}$ **d** $\begin{pmatrix}1\\4\end{pmatrix}$

e $\begin{pmatrix}11\\9\end{pmatrix}$ **f** $\begin{pmatrix}14\\20\end{pmatrix}$

15 $x = 1, -1$

16 $x = 1$

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